РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №4

дисциплина: Администрирование сетевых подсистем

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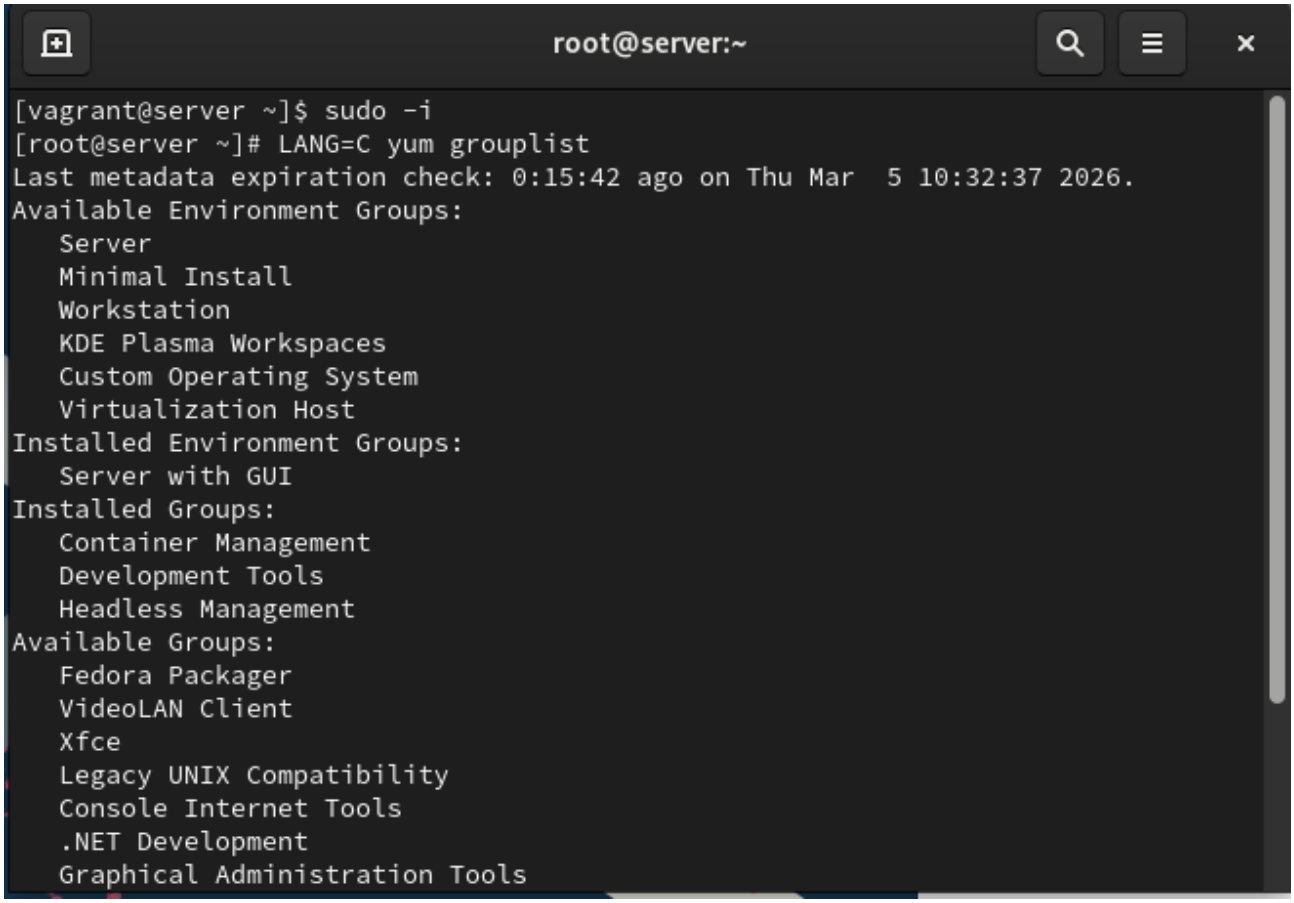
МОСКВА

2025 г.

Цель работы:

Целью данной работы является приобретение практических навыков по установке и базовому конфигурированию HTTP-сервера Apache.

Выполнение работы:



```
[root@server ~]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:16:28 ago on Thu 05 Mar 2026 10:32:37 AM UTC.
Dependencies resolved.
================================================================================
 Package                Arch       Version                    Repository     Size
================================================================================
Installing group/module packages:
 httpd                  x86_64     2.4.62-7.el9_7.3           appstream      45 k
 httpd-manual           noarch     2.4.62-7.el9_7.3           appstream     2.2 M
 mod_fcgid              x86_64     2.3.9-28.el9               appstream      74 k
 mod_ssl                x86_64     1:2.4.62-7.el9_7.3         appstream     110 k
Installing dependencies:
 apr                    x86_64     1.7.0-12.el9_3             appstream     122 k
 apr-util               x86_64     1.6.1-23.el9               appstream      94 k
 apr-util-bdb           x86_64     1.6.1-23.el9               appstream      12 k
 httpd-core             x86_64     2.4.62-7.el9_7.3           appstream     1.4 M
 httpd-filesystem       noarch     2.4.62-7.el9_7.3           appstream      12 k
 httpd-tools            x86_64     2.4.62-7.el9_7.3           appstream      79 k
 rocky-logos-httpd      noarch     90.16-1.el9                appstream      24 k
```

```
Complete!
[root@server ~]# cd /etc/httpd/conf и /etc/httpd/conf.d
-bash: cd: too many arguments
[root@server ~]# cd ..
[root@server /]# cd /etc/httpd/conf и /etc/httpd/conf.d
-bash: cd: too many arguments
[root@server /]# cd /etc/httpd/conf
[root@server conf]# ls
httpd.conf  magic
[root@server conf]# cd  /etc/httpd/conf.d
[root@server conf.d]# ls
autoindex.conf  manual.conf  rocky-snipolicy.conf  userdir.conf
fcgid.conf      README       ssl.conf              welcome.conf
[root@server conf.d]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client imap imaps pop3 pop3s smtp ssh
[root@server conf.d]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp am
qps apcupsd audit ausweisapp2 bacula bacula-client bareos-director bareos-fileda
emon bareos-storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-r
pc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit col
lectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dh
```

```
t imap imaps ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin
 kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-apiserver
kube-control-plane kube-control-plane-secure kube-controller-manager kube-contro
ller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure
kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-t
ls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix m
dns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql murmur mysq
l nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvp
n ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi
pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis re
dis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client sam
ba-dc sane sip sips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snm
ptrap spideroak-lansync spotify-sync squid ssdp ssh steam-streaming svdrp svn sy
ncthing syncthing-gui syncthing-relay synergy syslog syslog-tls telnet tentacle
tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpi
nator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discove
ry-tcp ws-discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp
-server zabbix-agent zabbix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=http
success
[root@server conf.d]# firewall-cmd --add-service=http --permanent
success
[root@server conf.d]#
```

```
root@server:~
root@server:/etc/httpd/conf.d        root@server:~
[vagrant@server ~]$ sudo -i
[root@server ~]# journalctl -x -f
Mar 05 11:29:02 server.user.net anacron[6866]: Job `cron.daily' terminated
Mar 05 11:29:02 server.user.net anacron[6866]: Normal exit (1 job run)
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable
 to get EDID for output
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable
 to get EDID for output
Mar 05 11:29:54 server.user.net gsd-color[5051]: unable to get EDID for xrandr-Virtual1: unable
 to get EDID for output
Mar 05 11:33:09 server.user.net systemd[3466]: Started VTE child process 7894 launched by gnome
-terminal-server process 5554.
    Subject: A start job for unit UNIT has finished successfully
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit UNIT has finished successfully.

    The job identifier is 589.
Mar 05 11:33:14 server.user.net sudo[7920]:  vagrant : TTY=pts/1 ; PWD=/root ; USER=root ; COMM
AND=/bin/bash
Mar 05 11:33:14 server.user.net sudo[7920]: pam_unix(sudo-i:session): session opened for user r
oot(uid=0) by vagrant(uid=1000)
Mar 05 11:33:14 server.user.net systemd[1]: Starting Hostname Service...
    Subject: A start job for unit systemd-hostnamed.service has begun execution
    Defined-By: systemd
    Support: https://wiki.rockylinux.org/rocky/support

    A start job for unit systemd-hostnamed.service has begun execution.
```

```
mountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3
 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pm
proxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
y-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rp
c-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp s
mtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ss
dp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-
tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server
warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
discovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabb
ix-server zerotier
[root@server conf.d]# firewall-cmd --add-service=http
success
[root@server conf.d]# firewall-cmd --add-service=http --permanent
success
[root@server conf.d]# systemctl enable httpd
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/sy
stem/httpd.service.
[root@server conf.d]# systemctl start httpd
[root@server conf.d]#
```

```
root@client:/

[vagrant@client ~]$ sudo -i
[root@client ~]# tail -f /var/log/httpd/error_log
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or direct
ory
tail: no files remaining
[root@client ~]# cd ..
[root@client /]# tail -f /var/log/httpd/error_log
tail: cannot open '/var/log/httpd/error_log' for reading: No such file or direct
ory
tail: no files remaining
[root@client /]#
```

```
sync elasticsearch etcd-client etcd-server finger foreman foreman-proxy freeipa-4 freeipa-ldap
freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-client ganglia-master git g
sd grafana gre high-availability http http3 https ident imap imaps ipfs ipp ipp-client ipsec i
c ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell
kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager k
be-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-
orker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network
llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh
ountd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3
nmea-0183 nrpe ntp nut openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pmcd pm
roxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter prox
-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rdp redis redis-sentinel rp
-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip sips slp smtp s
tp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ss
p ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-
ls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server
arpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-
iscovery-udp wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabb
x-server zerotier
root@server conf.d]# firewall-cmd --add-service=http
uccess
root@server conf.d]# firewall-cmd --add-service=http --permanent
uccess
root@server conf.d]# systemctl enable httpd
reated symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/sy
tem/httpd.service.
root@server conf.d]# systemctl start httpd
root@server conf.d]# tail -f /var/log/httpd/access_log
```

```
[root@server ~]# cd /etc/httpd/conf.d
[root@server conf.d]# touch server.user.net.conf
[root@server conf.d]# touch www.user.net.conf
[root@server conf.d]# gedit user.net

(gedit:8297): dconf-WARNING **: 11:49:18.228: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.245: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.957: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.962: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)

(gedit:8297): dconf-WARNING **: 11:49:18.982: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)
```

```
(gedit:8297): dconf-WARNING **: 11:50:07.511: failed to commit changes t
ute child process “dbus-launch” (No such file or directory)
[root@server conf.d]# cd /var/www/html
[root@server html]# mkdir server.user.net
[root@server html]# cd /var/www/html/server.user.net
[root@server server.user.net]# touch index.html
[root@server server.user.net]# gedit

(gedit:8325): dconf-WARNING **: 11:51:15.509: failed to commit changes t
ute child process “dbus-launch” (No such file or directory)

(gedit:8325): dconf-WARNING **: 11:51:15.512: failed to commit changes t
ute child process “dbus-launch” (No such file or directory)
```

Open ▾ | index.html — /var/www/html/server.user.net | Save | ≡ | ×

```
1 Welcome to the server.user.net server.
```

```
root@server:/var/www/html/www.user.net
root@server:/etc/httpd/c...   root@server:~   root@server:/var/www/h...
data::gedit-position not supported

(gedit:8398): dconf-WARNING **: 11:53:07.322: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)
[root@server server.user.net]# cd /var/www/html
[root@server html]# mkdir www.user.net
[root@server html]# cd /var/www/html/www.user.net
[root@server www.user.net]# chown -R apache:apache /var/www
[root@server www.user.net]# restorecon -vR /etc
restorecon -vR /var/named
restorecon -vR /var/www
Relabeled /etc/named from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:named_conf_t:
s0
Relabeled /etc/named/keys from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:named_co
nf_t:s0
Relabeled /etc/named/keys/dhcp_updater.key from unconfined_u:object_r:etc_t:s0 to unconfined_u:
object_r:named_conf_t:s0
Relabeled /etc/named.conf from unconfined_u:object_r:etc_t:s0 to unconfined_u:object_r:named_co
nf_t:s0
restorecon: lstat(/var/named) failed: No such file or directory
[root@server www.user.net]# restorecon -vR /etc
[root@server www.user.net]# restorecon -vR /var/named
restorecon: lstat(/var/named) failed: No such file or directory
[root@server www.user.net]# mkdir /var/named
[root@server www.user.net]# restorecon -vR /var/named
Relabeled /var/named from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:named_zone_t:
s0
[root@server www.user.net]# restorecon -vR /var/www
[root@server www.user.net]#
```

www.user.net

Rocky Linux Rocky Wiki Rocky Forums Rocky Mattermost Rocky Reddit

Firefox automatically sends some data to Mozilla so that we can improve your experience. Choose What I Share

Hmm. We're having trouble finding that site.

We can't connect to the server at www.user.net.

If you entered the right address, you can:

- Try again later
- Check your network connection
- Check that Firefox has permission to access the web (you might be connected but behind a firewall)

Open Save

*http.sh
/vagrant/provision/server

```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Install needed packages"
4 dnf -y groupinstall "Basic Web Server"
5 echo "Copy configuration files"
```

sh Tab Width: 8 Ln 5, Col 32 INS

```
root@server:/etc/httpd/c...   root@server:~   root@server:/vagrant/pr...
[root@server www.user.net]# restorecon -vR /var/named
Relabeled /var/named from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:named_zone_t:
s0
[root@server www.user.net]# restorecon -vR /var/www
[root@server www.user.net]# systemctl restart httpd
[root@server www.user.net]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server www.user.net]# cd .
[root@server www.user.net]# cd /
[root@server /]# cd /vagrant/provision/server
-bash: cd: /vagrant/provision/server: No such file or directory
[root@server /]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server /]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server /]# cp -R /etc/httpd/conf.d/*
/vagrant/provision/server/http/etc/httpd/conf.d/
cp: target '/etc/httpd/conf.d/www.user.net.conf' is not a directory
-bash: /vagrant/provision/server/http/etc/httpd/conf.d/: Is a directory
[root@server /]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/conf.d/
[root@server /]# cd /vagrant/provision/server/dns/
-bash: cd: /vagrant/provision/server/dns/: No such file or directory
[root@server /]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: cannot stat '/var/named/*': No such file or directory
[root@server /]# cd /vagrant/provision/server
[root@server server]# touch http.sh
[root@server server]# chmod +x http.sh
[root@server server]# getdit http.sh
bash: getdit: command not found...
[root@server server]# gedit http.sh
```

```
(gedit:8670): dconf-WARNING **: 12:04:22.291: failed to commit changes to dconf: Failed to exec
ute child process “dbus-launch” (No such file or directory)
[root@server server]# cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp: overwrite '/etc/httpd/conf.d/autoindex.conf'? yes
cp: overwrite '/etc/httpd/conf.d/fcgid.conf'? y
cp: overwrite '/etc/httpd/conf.d/manual.conf'? Y
cp: overwrite '/etc/httpd/conf.d/README'? Y
cp: overwrite '/etc/httpd/conf.d/rocky-snipolicy.conf'? all
cp: overwrite '/etc/httpd/conf.d/server.user.net.conf'?
cp: overwrite '/etc/httpd/conf.d/ssl.conf'?
cp: overwrite '/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/etc/httpd/conf.d/user.net'?
cp: overwrite '/etc/httpd/conf.d/welcome.conf'?
cp: overwrite '/etc/httpd/conf.d/www.user.net.conf'?
[root@server server]#
[root@server server]#
[root@server server]# cp -R /vagrant/provision/server/http/var/www/* /var/www
[root@server server]#
[root@server server]#
```

```
root@server:/vagrant/provision/server
root@server:/etc/httpd/c...   root@server:~   root@server:/vagrant/pr...
cp: overwrite '/etc/httpd/conf.d/manual.conf'? Y
cp: overwrite '/etc/httpd/conf.d/README'? Y
cp: overwrite '/etc/httpd/conf.d/rocky-snipolicy.conf'? all
cp: overwrite '/etc/httpd/conf.d/server.user.net.conf'?
cp: overwrite '/etc/httpd/conf.d/ssl.conf'?
cp: overwrite '/etc/httpd/conf.d/userdir.conf'?
cp: overwrite '/etc/httpd/conf.d/user.net'?
cp: overwrite '/etc/httpd/conf.d/welcome.conf'?
cp: overwrite '/etc/httpd/conf.d/www.user.net.conf'?
[root@server server]#
[root@server server]#
[root@server server]# cp -R /vagrant/provision/server/http/var/www/* /var/www
[root@server server]#
[root@server server]# chown -R apache:apache /var/www
[root@server server]# restorecon -vR /etc
[root@server server]# restorecon -vR /var/www
[root@server server]# echo "Configure firewall"
Configure firewall
[root@server server]# firewall-cmd --add-service=http
Warning: ALREADY_ENABLED: 'http' already in 'public'
success
[root@server server]# firewall-cmd --add-service=http --permanent
Warning: ALREADY_ENABLED: http
success
[root@server server]# echo "Start http service"
Start http service
[root@server server]# systemctl enable httpd
[root@server server]# systemctl start httpd
[root@server server]#
```

Вывод:

В ходе выполнения лабораторной работы были приобретены практические навыки по установке и базовому конфигурированию HTTP-сервера Apache.

Ответы на контрольные вопросы:

1. Через какой порт по умолчанию работает Apache? - **По умолчанию, Apache работает через порт 80 для HTTP и порт 443 для HTTPS.**
2. Под каким пользователем запускается Apache и к какой группе относится этот пользователь? - **Apache обычно запускается от имени пользователя www-data (или apache, в зависимости от дистрибутива) и относится к группе с тем же именем.**
3. Где располагаются лог-файлы веб-сервера? Что можно по ним отслеживать? - **Лог-файлы веб-сервера обычно располагаются в директории логов. Например, в Ubuntu логи Apache хранятся в /var/log/apache2/, а в CentOS - в /etc/httpd/logs/. Лог-файлы содержат информацию о запросах к серверу, ошибки, статусы запросов и другие события, что позволяет администраторам отслеживать активность и выявлять проблемы.**
4. Где по умолчанию содержится контент веб-серверов? - **Контент веб-серверов по умолчанию обычно находится в директории, называемой "DocumentRoot". Например, в Apache на Linux DocumentRoot по умолчанию установлен в /var/www/html/. В этой директории содержатся файлы, которые веб-сервер отдает при запросах.**
5. Каким образом реализуется виртуальный хостинг? Что он даёт? - **Виртуальный хостинг в Apache позволяет хостить несколько сайтов на одном сервере. Это можно сделать двумя способами:**

   Базовый виртуальный хостинг: Разные сайты обслуживаются на одном сервере с использованием разных директорий DocumentRoot.

Name-based (или Host-based) виртуальный хостинг: Разные сайты обслуживаются на одном IP-адресе, но на разных доменных именах. Это основывается на значении заголовка "Host" в HTTP-запросе, который используется для определения, какой виртуальный хост должен обработать запрос.

Виртуальный хостинг позволяет хозяину сервера размещать несколько сайтов на одном физическом сервере, управлять ими независимо, и предоставлять услуги хостинга для различных клиентов или проектов.